

GMM Clustering Analysis Report

Cluster Distribution

1 286688

0 284998

2 55354

Cluster Characteristics

sigai hour day_of_week month

GMM_Cluster

0 0.006517 5.436508 2.998940 5.772841

1 0.006986 17.538125 2.955286 5.846397

2 0.021022 11.522979 3.225512 7.540033

Interpretation of Results

Cluster Distribution

- **Cluster 1**: 286,688 data points
- **Cluster 0**: 284,998 data points
- **Cluster 2**: 55,354 data points

Cluster Characteristics

- **Cluster 0**:
 - Lower `sigai` values.
 - Mostly observed in the early morning hours (average hour: 5.44).
 - Occurs around the middle of the week (day 3, where 0 = Monday).
 - Most common in May.

- **Cluster 1**:

- Moderate `sigai` values.
- Mostly observed in the evening hours (average hour: 17.54).
- Similar day distribution as Cluster 0.
- Most common in May.

- **Cluster 2**:

- Higher `sigai` values.
- Mostly observed in the late morning hours (average hour: 11.52).
- Predominantly occurs around the middle of the week (day 3).
- Most common in July.

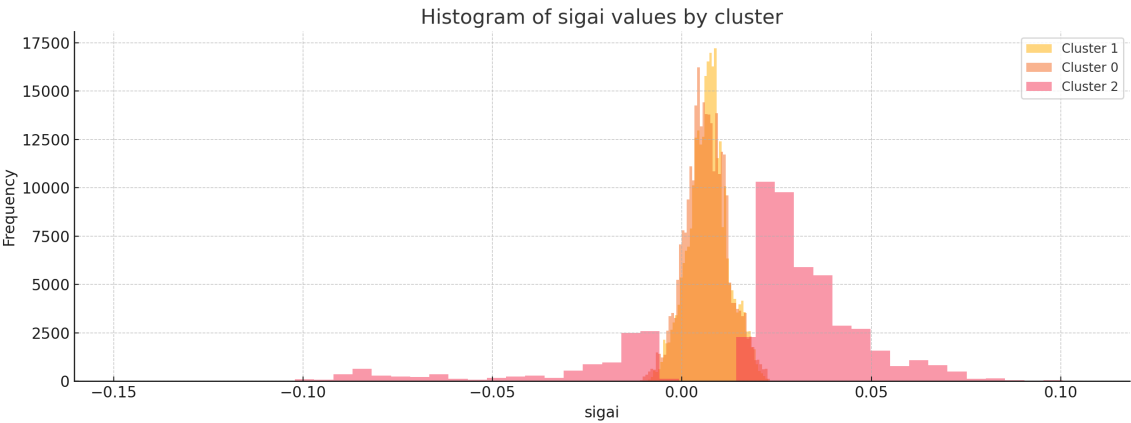
Visualizations

The visualizations included provide insights into the distribution and temporal patterns of `sigai` for each cluster:

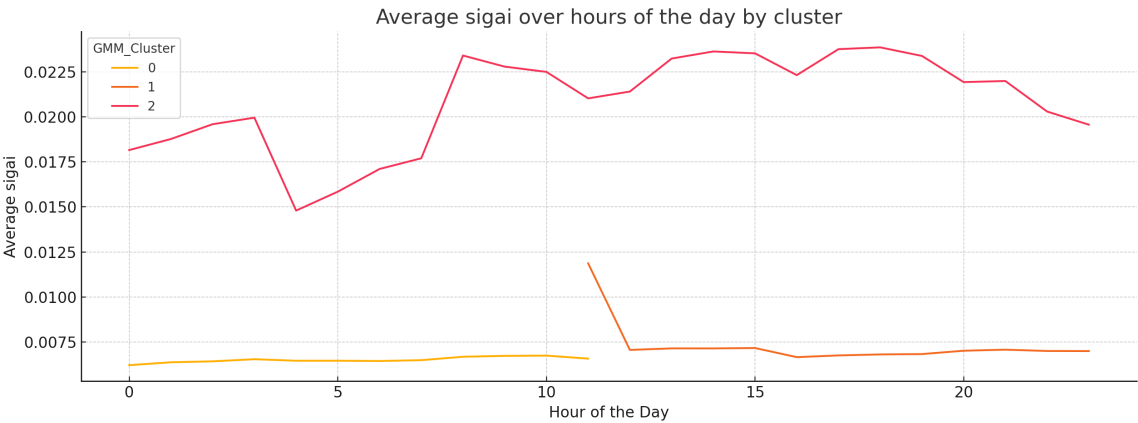
1. **Histogram of `sigai` values by cluster**: Shows the distribution of `sigai` within each cluster.
2. **Average `sigai` over hours of the day by cluster**: Illustrates how the average `sigai` changes throughout the day for each cluster.
3. **Average `sigai` over days of the week by cluster**: Demonstrates the weekly trends in `sigai` for each cluster.
4. **Average `sigai` over months by cluster**: Captures the monthly trends in `sigai` for each cluster.

These insights help to understand the patterns and characteristics of the clusters formed by the GMM algorithm. The `sigai` values vary significantly across different times of the day, days of the week, and months, showing distinct behaviors for each cluster. This analysis can be useful for identifying specific time-based trends and making data-driven decisions.

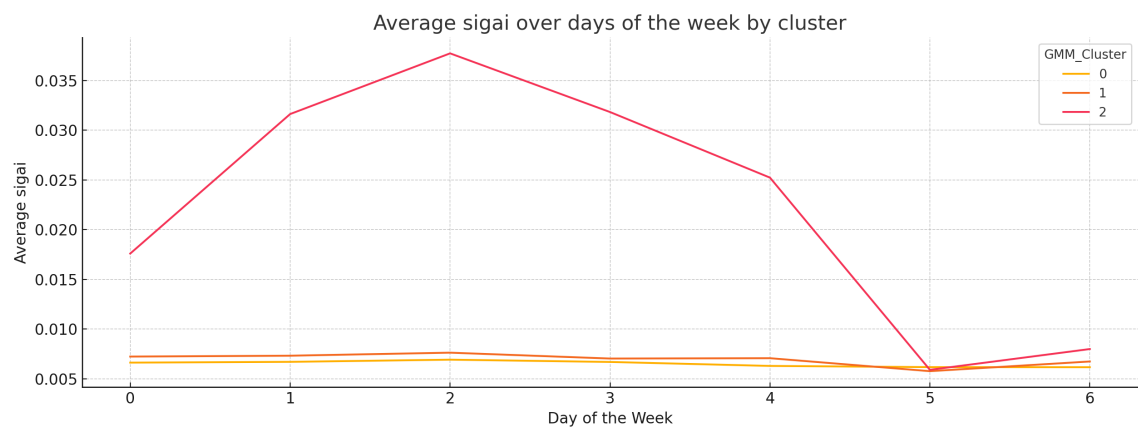
Histogram of sigai values by cluster



Average sigai over hours of the day by cluster



Average sigai over days of the week by cluster



Average sigai over months by cluster

